VIPS
Technical Campus
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IN PURSUIT OF PERFECTION

# COLLEGE OF ENGINEERING

# Fuels

# Liquid Fuels



- Extensively used in both industrial and domestic fields.
- The importance of liquid fuels is the fact that almost all combustion engines run on them.
- The largest source of liquid fuels is petroleum. The calorific value of petroleum is about 40000 kJ/kg.

## Petroleum

- Petroleum means ‘rock oil’ (In Latin, petra means rock and oleum means oil.)
- Also called mineral oil.
- It is a complex mixture of paraffinic, olefinic, and aromatic hydrocarbons with small quantities of organic compounds containing oxygen, nitrogen and sulphur.

# Refining of Petroleum



- Crude oil reaching the surface, generally consists of a mixture of solid, liquid and gaseous hydrocarbons containing sand and water.
- After the removal of dirt, water and much of the associated natural gas, the crude oil is separated into a no. of useful fractions by fractional distillation.
- The resultant fractions are then subjected to purification or conversion processes, known as refining of petroleum.

# Refining of Petroleum



**Demulsific-ation**

- The crude oil coming out from the well is in the form of a stable emulsion of oil and salt water. It is stabilized because of the presence of emulsifying agents.
- The demulsification is achieved by **Cottrell's process** in which the water is removed from the oil by electrical process.
- The crude oil is subjected to an electrical field, when droplets of colloidal water coalesce to form large drops which separate out from the oil.

**Removal of harmful impurities**

- Excessive salt content such as NaCl and $MgCl_2$ can corrode the refining equipment. These are removed by washing with water.
- The Sulphur compounds are removed by treating the oil with copper oxide. The copper sulphide so formed is separated by filtration.

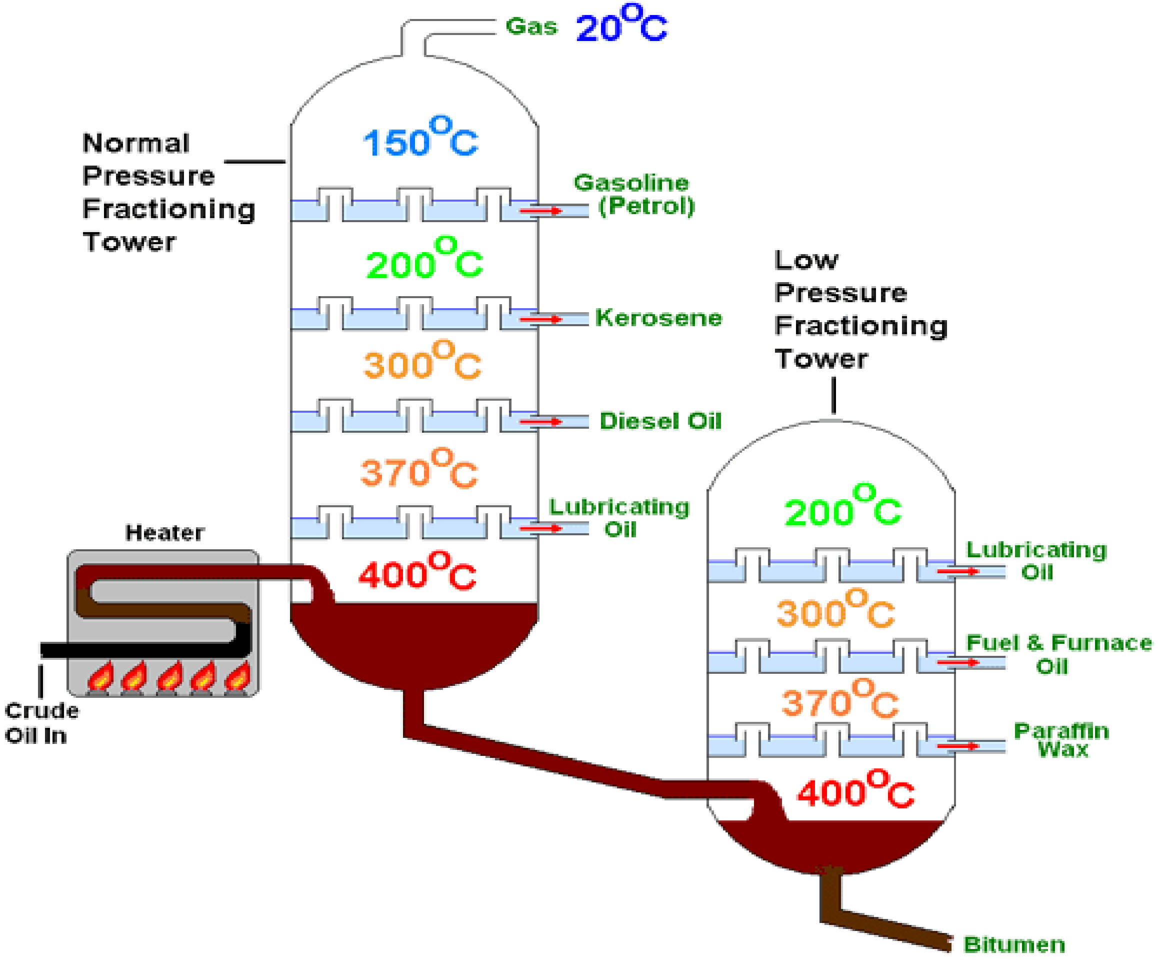
**Fractional Distillation**

- The Fractional distillation of petroleum is carried out in a specially designed tall fractionating tower or column made up of steel.

# Fractional Distillation



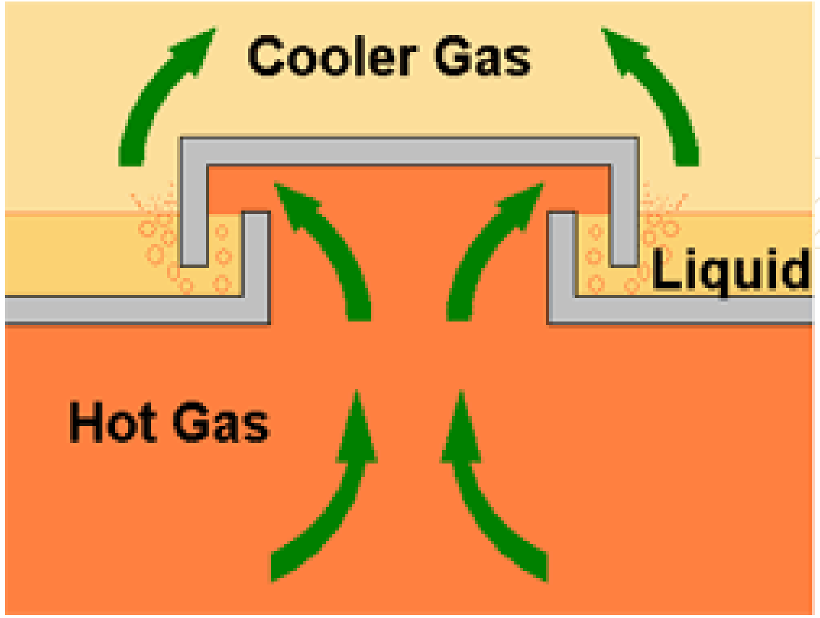
- The boiling point of a Hydrocarbon fraction is dependent on the length of the Carbon chain.
- Those fractions with shorter chains evaporate more easily than those with longer chains.
- In order to separate the different length chains in the crude mix, it is heated to a very high temperature.
- The temperature is set so that all those fractions with a Carbon chain length of 20 and below are evaporated from the crude mix. The temperature cannot be set higher than this as there is a risk that the lighter fractions will ignite.
- The remaining liquid, which is composed of only the heavier fractions, passes to a second location where it is heated to a similar temperature, but at lower pressure.



# Fractional Distillation



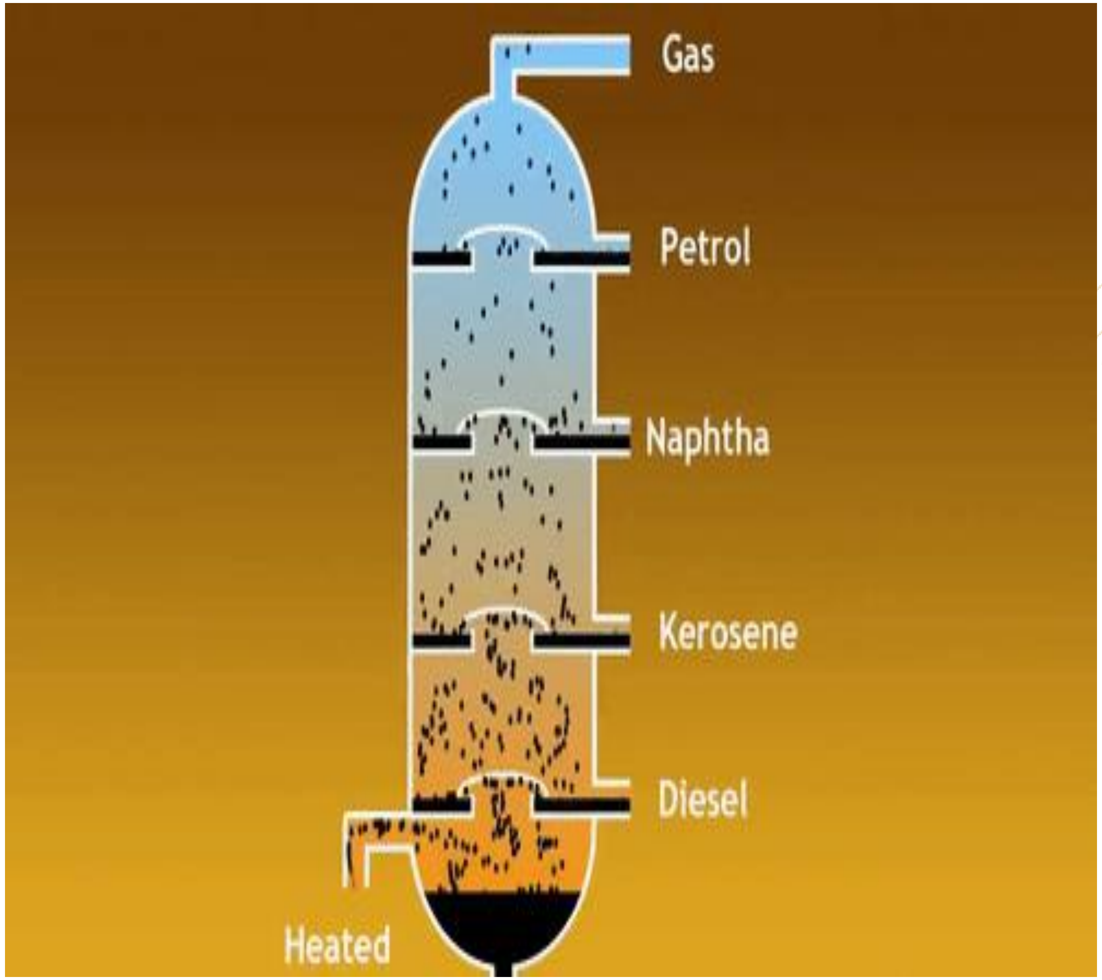
- The way the Distillation Tower works is by becoming progressively cooler from the base to the top.
- All the Hydrocarbon fractions start off in gas form, as they have been heated to that point. The gases then rise up the tower.
- The gas mixture then encounters a barrier through which there are only openings into the bubble caps. The gas mixture is then forced to go through a liquid before continuing upwards.
- The liquid in the first tray is at a cool enough temperature to get the heaviest gas fractions to condense into liquid form, while the lighter fractions stay gaseous.
- In this way the heaviest hydrocarbon fractions are separated out from the mixed gas.
- The remaining gas continues its journey up the tower until it reaches another barrier.



# Fractional Distillation



- This process continues until only the very lightest fractions, those of 1 to 4 Carbon atoms, are left. These stay in gas form and are collected at the top of the tower.



# Fractions obtained out of Crude Petroleum

| Fraction | Boiling point range (°C) | Number of C atoms | Uses |
|---|---|---|---|
| petroleum gas | < 40 | 1-4 | fuel / feedstock for petrochemicals |
| gasoline/naphtha | 40-170 | 5-10 | gasoline: fuel<br>naphtha: feedstock |
| kerosene | 170-250 | 10-16 | jet fuel<br>domestic heating |
| diesel | 250-350 | 11-18 | fuel for trucks/agricultural machinery |
| fuel oil | 350-450 | 20-25 | fuel for ships/industry |
| residue | > 450 | > 25 | lubricating oils/bitumen for roads |

# Cracking



- Of all the fractions obtained by fractional distillation of crude petroleum, gasoline is considered to be the most important fraction.
- The yield of this fraction is only 20% of the crude oil. The yield of heavier petroleum fraction is quite high.
- Therefore, heavier fractions are converted into more useful fraction, gasoline. This is achieved by a technique called **cracking.**
- Cracking is the process by which heavier fractions are converted into lighter fractions by the application of heat, with or without catalyst.
- Cracking involves the rupture of C-C and C-H bonds in the chains of high molecular weight hydrocarbons. Eg.

$$\underset{\text{Decane}}{C_{10}H_{22}} \rightarrow \underset{\text{n-pentane}}{C_5H_{12}} + \underset{\text{Pentene}}{C_5H_{10}}$$

$$C_8H_{18} \rightarrow C_5H_{12} + C_3H_6$$

# Process of Cracking



- Higher hydrocarbons are converted to lower hydrocarbons by C-C cleavage. The product obtained on cracking have low boiling points than the initial reactant.
- Formation of branched chain hydrocarbons takes place from straight chain alkanes.
- Unsaturated hydrocarbons are obtained from saturated hydrocarbons.
- Cyclization may takes place.

# Types of Cracking



- **Thermal Cracking:** When it takes place simply by the application of heat and pressure, the process is called thermal cracking.
- The heavy oils are subjected to high temperature and pressure, when the bigger hydrocarbons break down to give smaller molecules of paraffins, olefins etc.
- The thermal stability among the constituents of petroleum fractions increases as

  **Paraffins < naphthenes (alicyclics) < aromatics**

# Thermal Cracking: types



1. **Liquid Phase thermal cracking:** This method is used for the cracking of any type of oil (residues, heavy oil). The charge is kept in the liquid form by applying high pressures of the range 30-100 kg/cm$^2$ at a suitable temperature of 476-530$^o$C. The cracked products are separated in a fractionating column. The important fractions are: Cracked gasoline (30- 35%), Cracking gases (10-45%); Cracked fuel oil (50- 55%).
2. **Vapour phase thermal cracking:** By this method, only those oils which vapourize at low temperatures can be cracked. The petroleum fractions of low boiling range like kerosene oil, are heated at a temp of 670- 720 $^o$C under low pressure of 10-20 kg/cm$^2$.

# Thermal Cracking: Mechanism

*Initiation:*

$CH_3CH_3 \rightarrow CH_3^{\cdot} + CH_3^{\cdot}$

*Propagation:*

$CH_3CH_3 + CH_3^{\cdot} \rightarrow CH_4 + CH_3CH_2^{\cdot}$

$CH_3CH_2^{\cdot} \rightarrow CH_2 = CH_2 + H^{\cdot}$

$H^{\cdot} + CH_3CH_3 \rightarrow H_2 + CH_3CH_2^{\cdot}$

*Termination:*

$2CH_3CH_2^{\cdot} \rightarrow CH_3CH_2CH_2CH_3$

$CH_3CH_2^{\cdot} + H^{\cdot} \rightarrow CH_3CH_3$

# Catalytic Cracking : types



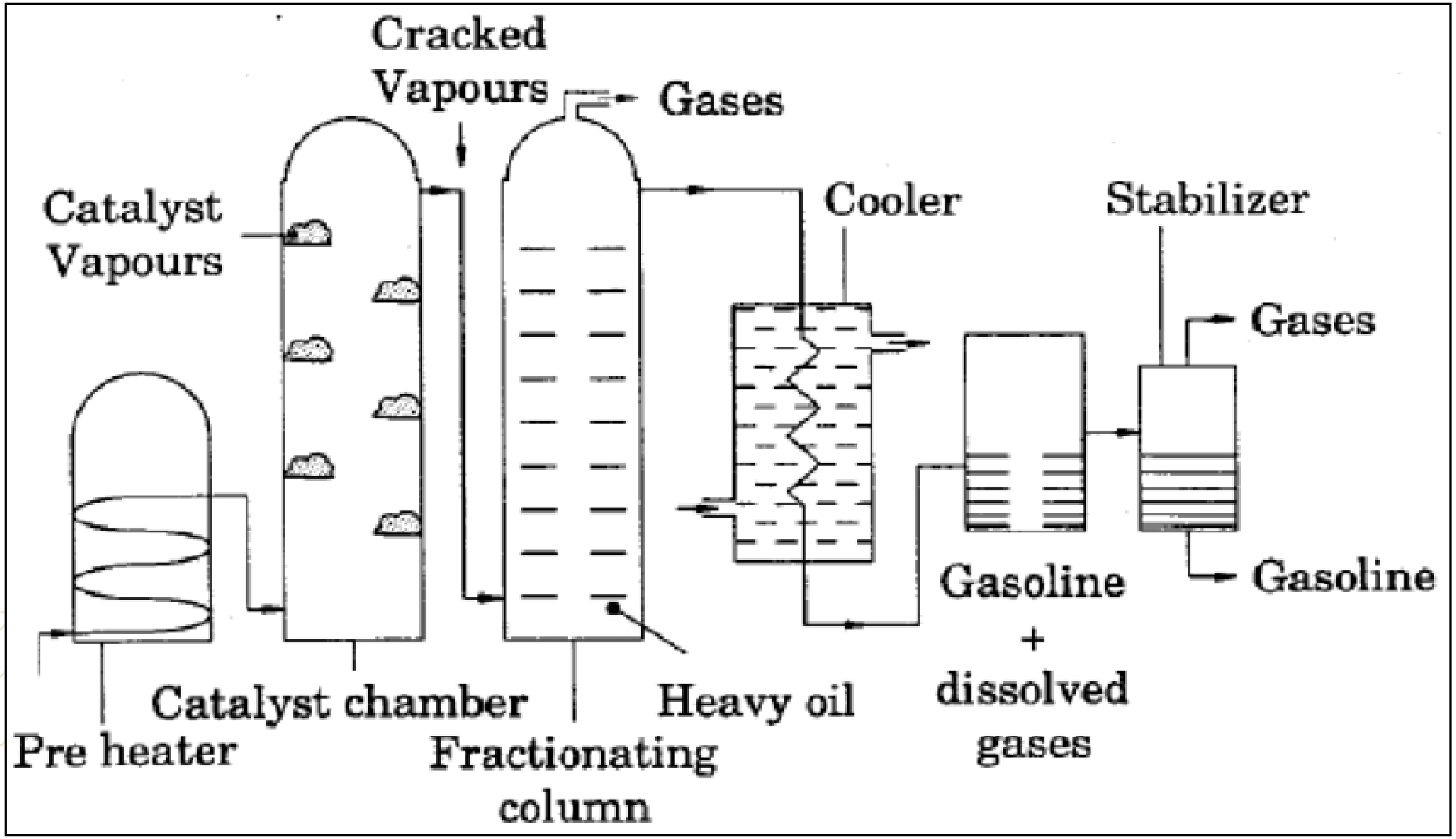
This type of cracking is brought about in the presence of a catalyst at much lower temperatures and pressures.

- The catalyst used is mainly a mixture of silica and alumina. Most recent catalyst used is zeolite. The quality and yield of gasoline is greatly improved by this method.
- The catalytic cracking is of two types:

1. Fixed-bed type catalytic cracking
2. Moving-bed Catalytic Cracking

# Fixed-bed Catalytic Cracking





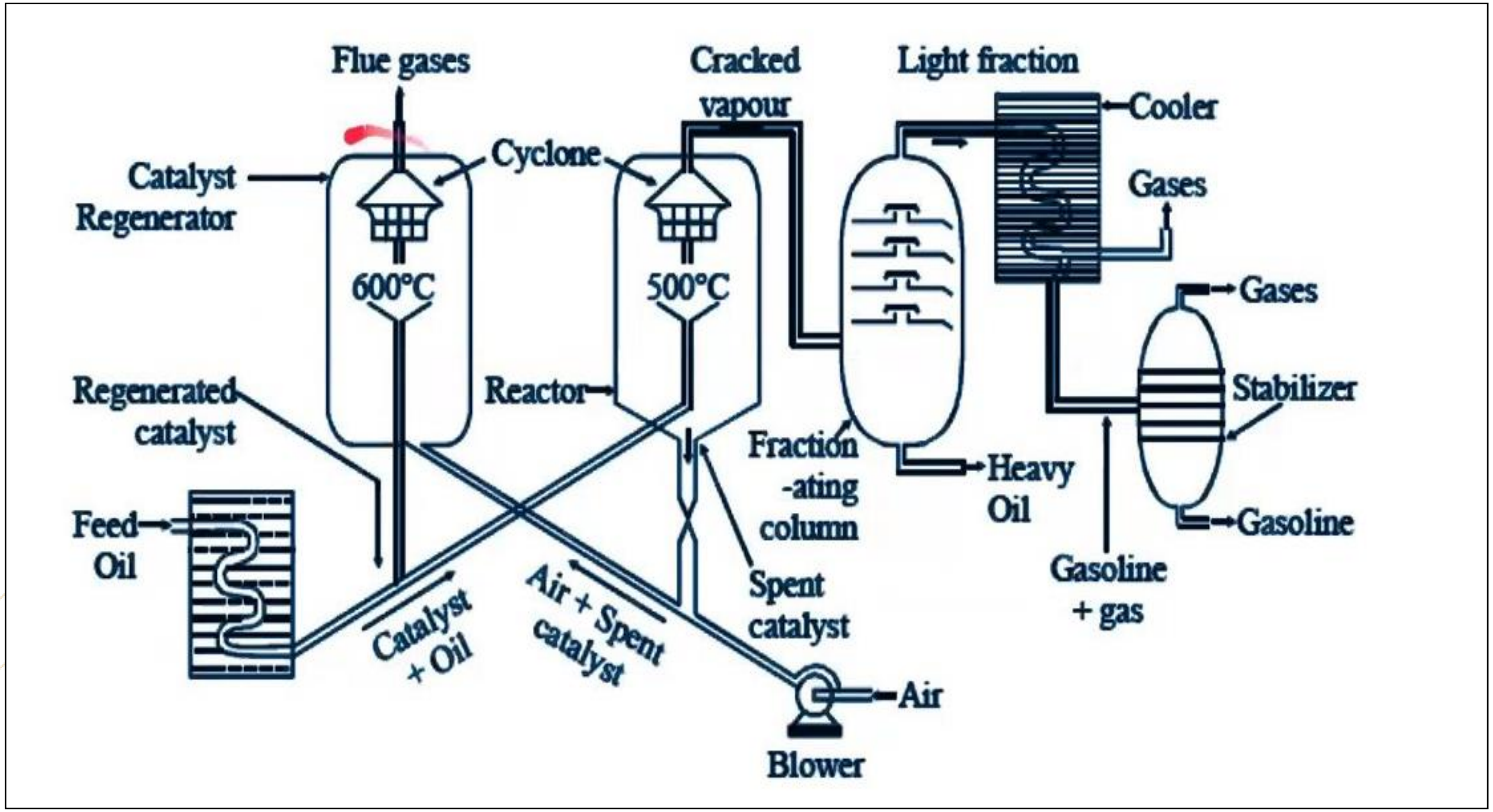

- ❖ The catalyst is in the form of granules or pellets. These pellets are used in the form of fixed beds in catalyst towers.
- ❖ The catalyst used is either silica-alumina gel or bauxite, mixed with clay and zirconium oxide.

# Fixed-bed Catalytic Cracking



- The heavy oil charge is heated in a pre-heater to cracking temperatures (420 – 450°C) and then forced through a catalyst chamber maintained at 425 – 450°C and 1.5 kg/cm$^2$ pressure.
- During their passage through the tower, about 40% of the charge is converted into low molecular weight hydrocarbons. About 2 – 4% carbon is formed, which gets adsorbed on the catalyst bed.
- The vapor produced is then passed through a fractionating column, where heavy oil fractions condense.
- The vapors are then led through a cooler, where some of the gases are condensed along – with gasoline and uncondensed gases move on.
- The gasoline containing some dissolved gases is then sent to a 'stabilizer', where the dissolved gases are removed and pure gasoline is obtained.
- The catalyst, after 8 to 10 hours, stops functioning, due to the deposition of black layer of carbon, formed during cracking.
- This is re-activated by burning off the deposited carbon. During the re-activation interval, the vapors are diverted through another catalyst chamber.

# Moving-bed Catalytic Cracking



# Moving-bed Catalytic Cracking



- Heated oil vapors go up in the reactor and catalyst comes down through the hopper which is the significance in the moving bed catalytic cracking process.
- After the cracking of vapors the spent catalyst is removed from the bottom. It is regenerated and sent again to the catalyst hopper through the elevators.
- The cracked vapors after the separation of dust separated go to the fractionators where gas oil is separated from vapors of gas and gasoline. Gas oil is withdrawn from the bottom.
- The gas and the gasoline vapors are condensed in the condenser and are separated.

## Advantage of moving bed catalytic cracking:

1. Better contact with the feed and catalyst, enabling uniform temperature and efficient transfer
2. Catalyst can be regenerated and used again for cracking

# Advantages of Catalytic cracking over Thermal cracking



- High temp and pressure are not required in the presence of a catalyst.
- The use of catalyst not only accelerates the cracking reactions but also introduces new reactions which considerably modify the yield and the nature of the products.
- The yield of the gasoline is higher.
- No external fuel is required for cracking.
- The process can be better controlled so desired products can be obtained.

- The product contains a very little amount of undesirable sulphur because a major portion of it escapes out as $H_2S$ gas, during cracking.
- Catalysts are selective in action and hence cracking of only high boiling fractions takes place.
- Coke forming materials are absorbed by the catalysts as soon as they are formed.

# GASEOUS FUEL



Gas fuels are the most convenient because they require the least amount of handling and are used in the simplest and most maintenance-free burner systems. The gaseous fuels are most preferred because of their ease of storage, transport, handling and ignition.

These are classified into two types:

1. Primary fuels. Ex:- Natural gas
2. Secondary fuels. ex: - LPG, Coal gas, producer gas, water gas, Oil gas

# Properties

Gaseous fuels are generally easier to handle and burn than are liquid or solid fuels.

Gaseous fuels in common use are Liquefied petroleum gases (LPG), Natural gas, producer gas, Coal gas, Water gas, Oil gas, blast furnace gas, coke oven gas etc. The calorific value of gaseous fuel is expressed in Kilocalories per normal cubic meter ($kCal/Nm^3$) i.e. at normal temperature ($20\ ^0C$) and pressure (760 mm Hg).

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# THANK YOU